PhAIL: Time-to-Success CDFs and Paired Tests for Real-Robot VLA Evaluation

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Abstract

We introduce PhAIL (Physical AI Leaderboard), a real-robot benchmark and evaluation methodology for vision-language-action (VLA) policies. Most published VLA evaluations report binary success rate, a reliability proxy that does not capture throughput; reporting them as separate scalars exposes the deeper problem that no single scalar uniquely encodes operational priority. We adopt the time-to-success CDF as a primitive that covers both axes jointly, with hard failures (drops outside the workspace, safety stops) absorbed as $T = \infty$ ghost events. Standard scalars (UPH, success rate at τ, MTBF/A, completion fraction) are projections of this CDF. Each task's CDF is anchored cell-wise to a teleoperated-human reference, yielding macro-averaged *Human-Relative Throughput* (HRT) as a dimensionless headline scalar. We then separate the two distinct jobs that a methodology must do: *reporting* (a scalar with bootstrap intervals) and *ranking* (a paired CDF-level test). On four frontier VLAs (OpenPI $\pi_{0.5}$ [$\pi_{0.5}$], GR00T N1.6 [GR00T N1], ACT [ACT], SmolVLA [SmolVLA]) evaluated on a Franka FR3 across four objects at $\sim$35 paired rollouts per (model, object) cell, the top two inference models are pairwise indistinguishable on every metric tested; the third is separable; the fourth clearly worst. Subsampling shows that a Kolmogorov-Smirnov test on the macro CDF detects a borderline ranking difference (GR00T vs. ACT) at N=20 paired rollouts and climbs sharply on the other close pairs, while binary success-rate-at-threshold and the integrated RMST scalar stay near the noise floor across the entire $N \leq 30$ range. The choice of test, not just the choice of N, determines what is resolvable. The dominant practice – single scalars at $N \leq 25$ per condition without confidence intervals or paired tests – is two orders of magnitude under-budgeted for the ranking claims it publishes. All policies were fine-tuned with their default open-source recipes; the contribution is methodological. Benchmark, dataset, and code: `https://phail.ai`.

1 Introduction

How do you tell whether one robot policy is better than another at a real task on real hardware? Most published VLA evaluations answer this with binary success rate at a fixed timeout. We argue that this answer is structurally insufficient, that the time-to-success CDF is the right primitive instead, and that once one adopts that primitive the methodology splits into two distinct jobs – *reporting* (a scalar projection of the CDF) and *ranking* (a paired test on the CDF itself) – which most current practice conflates and which behave very differently as a function of sample size.

Success rate is a reliability metric without a throughput axis. Reported as "did the policy complete the task within τ seconds?", it is a usable proxy for reliability: with τ generous enough that an in-distribution policy succeeds, the metric tells you whether the policy is reliable. But reliability is one operational property and throughput is a different one. A 90% success rate at τ=30 seconds is a deployable policy; the same 90% at τ=300 seconds is not. The standard fix is to add a throughput metric (UPH, cycle time, completion fraction) alongside the success rate. That works, but it surfaces the deeper question: when the two scalars rank the policies differently, which do you publish? The answer is task-dependent and not interchangeable, and the choice is rarely disclosed.

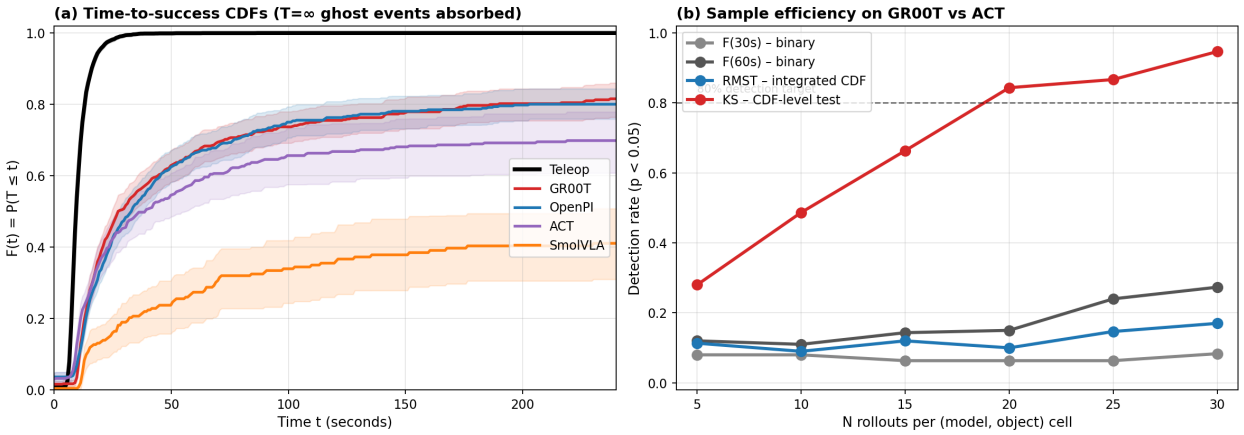


Figure 1: **Time-to-success CDFs and metric sample-efficiency on the PhAIL benchmark. (a)** Pooled time-to-success CDFs $F(t) = P(T \leq t)$ for four frontier VLAs and a teleoperated reference, with hard failures absorbed as $T = \infty$ ghost events. The asymptote below $F = 1$ equals the hard-failure rate; the curve's position in time encodes throughput (the median of T is where F crosses 0.5, and UPH $\propto 1/E[T]$). Reliability and throughput live on the same axis; every scalar reported in the field (success rate at threshold, UPH, MTBF/A, completion fraction) is a projection of this object. **(b)** Detection rate vs. subsample size N on the borderline pair GR00T vs. ACT. *Detection rate* is the fraction of bootstrap subsamples (200 outer trials, 100-rep inner episode-clustered bootstrap) that yield $p < 0.05$ on the difference of each candidate metric, macro-averaged across objects (§4.2). A Kolmogorov-Smirnov test on the macro CDF reaches 80% detection at N=20 paired rollouts per (model, object) cell; binary success-rate-at-threshold ($F(30\,\text{s})$, $F(60\,\text{s})$) and the integrated RMST scalar never cross the 80% line within $N \leq 30$. The choice of test, not just the choice of N, determines what is resolvable.

Time-to-success T covers both axes. Let T be the wall-clock time the policy needs to complete one operation of the task. A short T is fast *and* reliable; a long T is slow; an undefined T (item lost outside the workspace, safety stop fired) is a hard failure that can never recover. We extend the standard time-to-event apparatus by absorbing such hard failures as $T = \infty$ ghost events: an operation that physically cannot complete contributes one $(T, E) = (\infty, 1)$ observation rather than another censored tail. The resulting CDF $F(t) = P(T \leq t)$ jointly carries reliability (its asymptote below 1, equal to the hard-failure rate) and throughput (the median of T where F crosses 0.5, and UPH $\propto 1/E[T \mid T < \infty]$). Figure 1 shows the four frontier VLAs and the teleop reference under this primitive. UPH, success rate at τ, MTBF/A, and completion fraction are all projections of F. (Our task happens to consist of many operations per episode, which we exploit for sample size; the framework applies just as well to single-operation episodes.)

No single scalar projection uniquely encodes operational priority. Once F is the primitive, the natural next step is to derive a single scalar to rank by. We show empirically that this is structurally fragile: pairwise stochastic dominance against the teleoperated reference (AUC $= P(T_{\text{model}} < T_{\text{ref}})$) and integrated metrics (restricted-mean survival time, RMST) rank the same four policies in different orders on our data. The cause is not that AUC has an inherent bias; it is that the teleop reference is so much faster than any of the evaluated policies that the AUC integral is dominated by short-time behaviour, where models with sharp early-success peaks win, while integrated metrics weight all of $[0, \tau]$ equally. The takeaway is not "choose the right scalar" but "report the distribution and disclose the aggregation."

The choice of test, not just the choice of N, determines what is resolvable. Reporting a scalar with confidence intervals is one job; testing whether two policies differ on the underlying distribution is another. We treat them as separate. §4.2 shows that on borderline model pairs in our data, a Kolmogorov-Smirnov (KS) test on the macro CDF reaches 80% detection at N=20–30 paired rollouts per (model, object) cell where binary success-rate-at-threshold and the integrated RMST scalar do not within budget. The

integrator that makes RMST interpretable is the same step that smears the where-do-the-curves-separate information that a sup-norm test exploits. *Scalars are for reporting; tests are for ranking.* Conflating the two wastes statistical power.

Contributions.

1. **A canonical primitive.** The time-to-success CDF with hard failures absorbed as $T = \infty$ ghost events. Standard scalars are projections of it; we propose macro-averaged *Human-Relative Throughput* (HRT) as a dimensionless headline scalar grounded in a same-fixture teleoperated reference.
2. **Aggregation-rule disagreement as a structural finding.** Same data, principled aggregation rules, opposite rankings. No single scalar uniquely encodes operational priority.
3. **Sample-efficiency from the right test, not the right scalar.** KS on the CDF detects ranking differences on borderline pairs at N=20–30 where every binary baseline fails within budget; RMST as a scalar gives essentially no efficiency advantage over $F(60\,\text{s})$. The two-jobs framing – scalar for reporting, test for ranking – is the central methodological recommendation.
4. **An open benchmark and reference implementation.** PhAIL: four frontier VLAs, four trained objects, ∼750 episodes including 372 teleop reference trajectories, full operations logs, synchronized video, and an open-source evaluation framework that implements the methodology end-to-end.

Scope. The contribution is the comparison primitive, the ranking procedure, the power budget that backs ranking claims, and the open framework that makes new claims auditable. The per-model rankings are illustrative of what the methodology can resolve at a given depth, not architectural pronouncements; we used each model's default published fine-tuning recipe (§3.2). The PhAIL leaderboard is open to submissions (§5.5).

2 Time-to-Success CDF as Evaluation Primitive

The methodology has three components: the random variable and its CDF (§2.1), the reporting layer (scalars and visualizations derived from the CDF, §2.2), and the ranking layer (paired tests on the CDF, §2.3). The strict separation between reporting and ranking is the methodological core.

2.1 The Random Variable T

For a single rollout episode with multiple operations to perform, we extract a sequence of times-to-success T_i and event indicators $E_i \in \{0, 1\}$. A successful placement contributes T_i equal to the elapsed time since the previous successful placement (or since episode start, for the first), with $E_i = 1$. If the episode ends with operations remaining (operator stopped, timed out, model stalled), the unfinished attempt contributes one right-censored pair $(T_{\text{tail}}, 0)$ where T_{tail} = duration − last placement time; $E = 0$ means we know $T > T_{\text{tail}}$ but do not observe T.

Hard failures as $T = \infty$ ghost events. Some operations end in states the policy cannot recover from. The clearest case is an item dropped outside the workspace, where the item is physically gone from the scene. Items dropped within the workspace but outside both totes (e.g., on the table) are a softer category: in principle the policy can pick them back up, and we observed at least one rollout where this happened. We count those as failures only if they are still uncollected at episode end: as many failures as there are dropped-and-not-recovered items at the time the rollout terminates. A safety-stop termination is a third class: the rollout ends because the operator (or built-in safety reflexes) stopped it, and the operation that triggered the stop contributes one ghost event. We absorb all of these as $(T = \infty, E = 1)$ ghost events: the operation *did* terminate (so $E = 1$, not censored), but at $T = \infty$ rather than at any finite time. The CDF carries this as an asymptote below $F = 1$; the asymptote's height equals the hard-failure rate. Stalls and timeouts are explicitly *not* ghost events: those are slow-but-recoverable runs that belong in the censored tail, where they correctly indicate "the policy was working when we stopped watching it."

Multi-placements. Occasionally a single grasp deposits several items in one motion (e.g., when a cluster of small batteries comes loose together). Our annotation tooling records each successful placement individually with a timestamp; when annotators marked multiple placements within a fraction of a second, we cluster placements within $\Delta t < 0.3\,\text{s}$ into one multi-placement event. The first item in the cluster contributes its inter-arrival time T_i to the CDF; subsequent items contribute $T = 0$ with $E = 1$, since they were placed essentially simultaneously. The inter-arrival anchor advances by the cluster's first timestamp only. This is an annotation-pipeline detail rather than a property of the underlying primitive: the tooling could have recorded simultaneous timestamps directly, in which case there would be no Δt threshold to set.

The CDF $F(t) = P(T \le t)$ is estimated by the Kaplan-Meier product-limit estimator [Kaplan and Meier] on the (T, E) collection. Confidence intervals on derived scalars (RMST, HRT, $F(\tau)$ at fixed thresholds) are reported via 95% episode-clustered bootstrap [Efron and Tibshirani]; pointwise log-log intervals using the Greenwood [Greenwood] variance estimator are equivalent and described in [Kalbfleisch and Prentice]. Episodes with zero successes contribute a single right-censored observation of length τ_{episode}, the per-rollout time budget after which an unfinished operation is censored (set by the protocol; §3.2); we never drop them. The $T = \infty$ atom is handled directly by the estimator (a discrete jump at the right edge); for the integrated-metric calculation in §2.2 we set $T_{\text{ghost}} \to \tau$, the integration cap, so a hard failure inflates RMST by exactly τ minus the time the model would have spent on that operation if successful.

We considered a (per-attempt success probability, per-attempt cycle time) framing and abandoned it. "Attempt" is not cleanly defined even for pick-and-place: should an air-grasp count, or only a closed gripper that lifts the object? does a drop-and-retry count as one attempt or two? Two readers can derive different success rates from the same video by drawing the boundary differently. The time-to-success T avoids the question entirely – the only thing measured is the wall clock until the next success.

2.2 Reporting: Scalars and Visualizations from the CDF

The CDF is the lossless representation. Every standard scalar in published VLA evaluations is a projection of it (Table 1); reporting any single one without the CDF discards information.

Scalar	Functional of $F(t)$
Success rate at τ	$F(\tau)$
Median time-to-success	$\inf\{t : F(t) \ge 0.5\}$
RMST(τ) (restricted-mean survival time)	$\int_0^\tau \big(1 - F(t)\big)\, dt$
UPH (units per hour)	$3600/E[T \mid T < \tau_{\text{episode}}] \propto 1/\text{RMST}(\tau_{\text{episode}})$
Completion fraction	$F(\tau_{\text{episode}})$
MTBF/A (mean time between failures/assists)	$\text{RMST}(\tau_{\text{episode}})/\big(1 - F(\tau_{\text{episode}})\big)$

Table 1: Standard scalar metrics expressed as functionals of the time-to-success CDF $F(t)$. The per-rollout time budget τ_{episode} is the protocol's timeout (§3.2); $1 - F(\tau_{\text{episode}})$ is the hard-failure rate (operations that did not complete within budget, including $T = \infty$ ghost events).

Headline scalar: Human-Relative Throughput (HRT). For each (model, object) cell we compute

$$\text{HRT}(m, o) = \frac{\text{RMST}_{\text{ref},\,o}(\tau)}{\text{RMST}_{m,o}(\tau)}, \qquad \tau = 240\,\text{s},$$

the cell-wise ratio of teleoperated-reference to model RMST, reported as a percentage, and macro-aggregated across objects with equal weights. Three properties make this the right scalar to report. (i) It inherits the CDF's joint speed-and-completion property via $T = \infty$ inflating $\text{RMST}_{\text{model}}$. (ii) It is grounded in operator practice (UPH-equivalent against a same-fixture human reference, so embodiment confounds cancel). (iii) Cell-wise normalization absorbs object-difficulty differences – a hard object slows teleop and model proportionally – yielding a dimensionless quantity comparable across heterogeneous tasks. Cross-deployment comparison (different operators, different rooms, different teleop pacing) reduces to comparing ratios, not absolute throughputs.

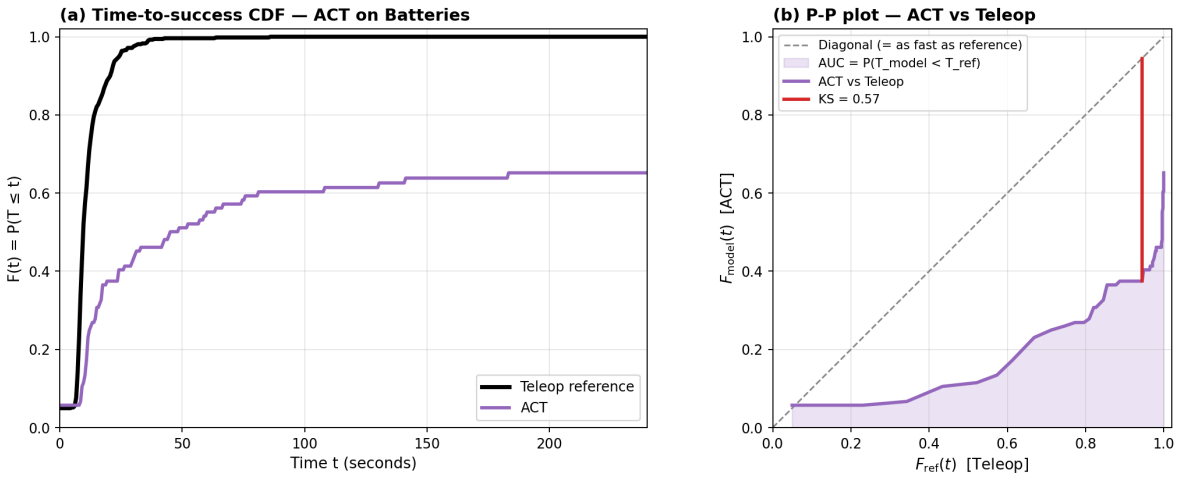


Figure 2: One (model, object) cell (ACT on Batteries) illustrating the reporting layer: time-to-success CDFs (left) and the corresponding P-P plot (right). **Left:** model and teleoperated-reference CDFs on the same axis; the asymptote below $F = 1$ on the model curve is the hard-failure rate. **Right:** $\big(F_{\text{ref}}(t),\, F_{\text{model}}(t)\big)$ traced parametrically as t sweeps. Diagonal = as fast as the reference; below-diagonal = slower. The shaded area under the curve is the AUC $P(T_{\text{model}} < T_{\text{ref}})$; the maximum vertical distance from the curve to the diagonal (red segment) is the Kolmogorov-Smirnov distance.

Visualization: P-P plots. For a (model, object) cell, the P-P plot traces $\big(F_{\text{ref}}(t),\, F_{\text{model}}(t)\big)$ parametrically as t sweeps from 0 to τ. The diagonal is "as fast as the reference"; below-diagonal is slower; above-diagonal is faster. Two derived properties give the P-P curve its analytic value. The area under the curve equals $P(T_{\text{model}} < T_{\text{ref}})$, the pairwise stochastic-dominance probability (AUC). The maximum vertical distance from the curve to the diagonal equals the Kolmogorov-Smirnov distance between the two CDFs (§2.3); the P-P plot is therefore the natural visualization for both. Figure 2 shows one (model, object) cell's CDF and the corresponding P-P plot side by side. P-P plots make distributional shape legible (saturation, early-success peaks, tail behavior); we use them descriptively in §4.3.

Confidence intervals. All reporting scalars and CDF curves carry 95% paired episode-clustered bootstrap intervals [Efron and Tibshirani]. We resample whole episodes (not operations within episodes) because operations within an episode are correlated by the rollout's shared visual conditions and policy state; per-operation bootstrap produces fake-tight intervals.

2.3 Ranking: Paired Tests on the CDF

Reporting a scalar with a CI is one job; testing whether two policies' underlying time-to-success distributions differ is a different job, and we treat it as such. Our primary test for ranking claims is the two-sample Kolmogorov-Smirnov statistic [Smirnov]

$$D = \sup_t \big|F_a(t) - F_b(t)\big|,$$

computed per-object on the macro-aggregated CDF, with p-values from a paired episode-clustered bootstrap [Efron and Tibshirani] that pools the two policies' subsamples per object under $H_0 : F_a = F_b$. Geometrically, D is the maximum vertical distance from the P-P plot of F_a versus F_b to the diagonal – the P-P visualization in §2.2 (Figure 2) reads off both AUC and KS by inspection. As a sanity-check secondary test we report stratified pooled-across-objects logrank [Mantel] with Bonferroni correction over the model pairs.

The two-jobs framing – scalar for reporting, KS for ranking – is non-trivial. §4.2 demonstrates empirically that RMST-as-scalar is not measurably more sample-efficient than $F(60\,\text{s})$ for *ranking*, while KS-as-test materially is. The information lives in *where the curves separate*; an integrator smears that information, a sup-norm test catches it.

3 PhAIL: Design, Protocol, and Power Budget

3.1 Platform and Task

The evaluation station uses a DROID-style configuration [DROID]: a Franka Research 3 with a Robotiq 2F-85 parallel-jaw gripper, plus an over-the-shoulder external RGB camera and a wrist-mounted RGB camera. The first task is bin-to-bin order picking: items move one at a time from an inbound (transparent) tote to an outbound (grey) tote across four object types spanning a useful slice of physical regimes – wooden spoons (rigid, elongated, multi-grasp), towels (deformable), scissors (articulated, metallic), and batteries (small, rigid). Bin-to-bin is deliberately not the most complex task we could put VLAs through; it is the simplest task with unambiguous success/intervention semantics that still carries clear commercial signal (millions of picks per day across warehousing and fulfilment). The four frontier VLAs we evaluate were trained for substantially more elaborate behaviour than this; the rankings reported below say something specific about how they perform on bin-to-bin under default fine-tuning recipes, not about their architectural ceilings on harder tasks. Subsequent PhAIL releases will add new tasks (including insertion and small-part assembly); the methodology is unchanged across additions. Figure 3 shows the platform during a single rollout: synchronized exterior and wrist video, 3D end-effector trajectory, and telemetry traces. Every evaluated rollout is published with these artifacts so any reader can audit a specific time-to-success annotation against the raw video and joint state. The benchmark, evaluation framework, dataset, and per-rollout artifacts are released at `https://phail.ai`.

3.2 Models, Fine-Tuning, and Protocol

We evaluate four publicly available VLAs: **OpenPI** $\pi_{0.5}$ [$\pi_{0.5}$], a 3B-parameter VLA with FAST action tokenization atop flow-matching; **NVIDIA GR00T N1.6** [GR00T N1], a 3B-parameter VLA combining a Cosmos-Reason VLM with a 32-layer diffusion transformer action head; **ACT** [ACT], the Action Chunking Transformer trained from scratch, a CVAE that predicts an action chunk; and **SmolVLA** [SmolVLA], a 450M-parameter VLA from the Hugging Face / LeRobot project. No model demonstrates zero-shot capability on this task, so fine-tuning is a prerequisite. We collected and publish 352 teleoperated demonstrations ($\sim$13 hours of robot operation; Appendix E). All four models were fine-tuned on the same dataset using each repository's default published recipe; we did not perform per-model hyperparameter search or recipe adaptation. **This is a deliberate methodological constraint, not best practice for production deployment;** a per-model recipe optimization could shift the rankings of §4.

Each rollout has a time cap of 30 s per item ($\sim$10$\times$ the $\sim$2.7 s/item teleop pace). Evaluation is blind: the scheduler randomly selects which model checkpoint runs next and the operator does not know which is running. The operator's only intervention is triggering a safety stop. After each rollout the operator records target/source counts, items lost outside both bins, and outcome label $\in$ {Success, Stalled, Safety, Ran_out_of_time}. Every successful placement is annotated post-hoc by a second annotator from the rollout video, yielding the (T_i, E_i) sequence per episode (Appendix H).

3.3 Cohort and Power Budget

After object/model filtering, $\sim$950 episodes enter the analyses; 386 are teleoperated reference rollouts. Annotations come from manual review of rollout video augmented by automated success detection (Appendix H); both label streams are released alongside the data.

How many rollouts does a ranking claim actually need? Two basic calculations, both standard, give the order-of-magnitude. First, if a paper wants to publish a confidence interval of width ± 5 percentage points around a single binary success rate, the Wilson formula needs roughly $N \approx 380$ rollouts per arm. Second, if a paper wants to detect a 5-percentage-point difference between two policies on a paired comparison with 80% power and $\alpha = 0.05$, the McNemar test needs roughly $N \approx 1280$ paired rollouts per condition (the range 600–1500 in this paper reflects different assumptions about how correlated the paired outcomes are; we use a discordance rate of 0.2 for the central estimate).

Both numbers are far above the field's modal $N = 10$–20 per condition, and both reach into the low thousands once the effects to resolve are realistic frontier-VLA gaps. Our own $N \approx 35$ per cell is 2–3$\times$

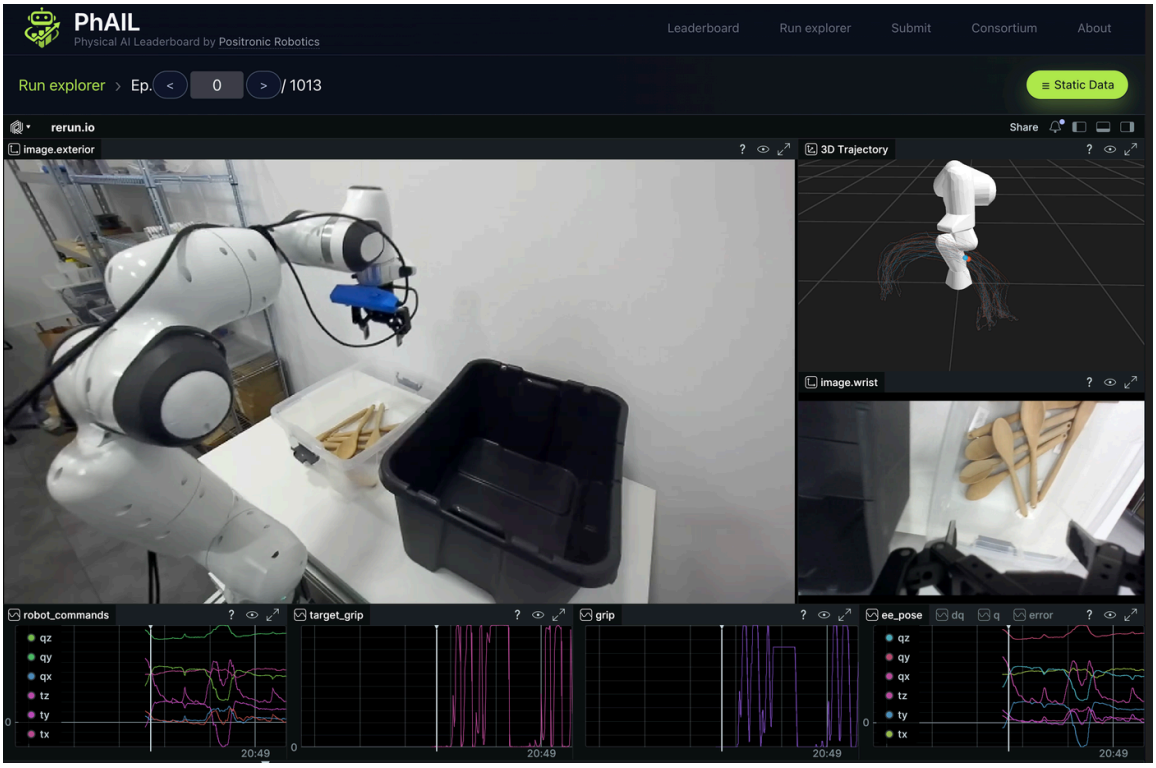


Figure 3: PhAIL platform during a rollout, visualized in the run-explorer interface (built on Rerun [Rerun]): synchronized exterior and wrist video, 3D end-effector trajectory, and telemetry traces (commanded action, gripper target, measured gripper). The same hardware operates the teleoperated reference and every evaluated VLA via the open-source *positronic* framework [Positronic], which exposes each model's observation/action contract behind a common station interface: the platform handles synchronization, action streaming, and safety checks uniformly, while each model contributes only its weights and the architecture-specific code that maps observations to action chunks. Released artifacts include the video streams, joint-state logs, automated event labels, and crowd-sourced human annotations of place timestamps and intervention counts.

the field median but still well below either threshold; §4.2 translates this analytical picture into empirical detection-rate-vs-N curves on our data, and reaches the same conclusion by a different route.

4 Results

All numbers below use $\tau = 240\,\text{s}$ for RMST, $n_{\text{boot}} = 1000$ for episode-clustered bootstrap CIs, seed 0. Per-(model, object) cell sizes are 30–45 episodes for OpenPI, GR00T, and ACT; 24–29 for SmolVLA.

4.1 Headline Ranking

GR00T and OpenPI are within 0.5 percentage points of HRT, well inside any reasonable noise floor at this N. No inference model exceeds 19% HRT on any single object (per-cell heatmap in Appendix B). Even the best frontier VLA is roughly 7× slower than a teleoperator on the same fixture.

The top two inference models cluster as statistically indistinguishable; ACT is separable from GR00T and borderline from OpenPI; SmolVLA is distinguishably worst. **At $N \approx 35$/cell – already 2–3× the field-modal N – the field's modal N cannot rank the top three.**

Model	RMST (s)	HRT (%)	Intervention rate	Episodes
Robot teleoperated by human	10.5	100.0	—	392
NVIDIA GR00T N1.6	78.3	**13.5**	2.3%	171
Physical Intelligence Open $\pi_{0.5}$	81.1	13.0	4.5%	177
Action Chunking Transformer	98.2	10.7	0.7%	148
Hugging Face SmolVLA	166.9	6.3	16.3%	123

Table 2: Headline ranking ($\tau = 240\,\text{s}$, $N \approx 35$/cell). HRT is the cell-wise reference-to-model RMST ratio, macro-averaged across four trained objects. Intervention rate is (safety stops + stalls) / total episodes (lower bound; drops outside the workspace not included).

Pair	p (raw)	p (Bonferroni)	Significance
OpenPI vs. SmolVLA	$<10^{-15}$	$<10^{-14}$	***
GR00T vs. SmolVLA	$<10^{-15}$	$<10^{-14}$	***
ACT vs. SmolVLA	3.3×10^{-16}	2.0×10^{-15}	***
GR00T vs. ACT	2.3×10^{-3}	1.4×10^{-2}	*
OpenPI vs. ACT	1.1×10^{-2}	6.7×10^{-2}	borderline
OpenPI vs. GR00T	4.5×10^{-1}	1.00	n.s.

Table 3: Pooled-across-objects logrank, Bonferroni-corrected over the 6 model-pairs.

What the field's standard methodology would have said. The binary success leader at the modal $\tau = 30$ seconds is GR00T (0.51), with OpenPI (0.46) and ACT (0.44) close behind. Switch the scalar to AUC-vs-teleop and the leader is *ACT* (0.138 vs. 0.101 for OpenPI and 0.096 for GR00T; §4.3). Switch again to integrated RMST and the leader is GR00T with OpenPI a hair behind, ACT a clear third. Same data, three principled scalar choices, three different top-1s. Table 4 below shows that at $N \leq 30$ paired per cell no binary-threshold metric can distinguish the close pairs on the underlying CDF: a reader of a field-standard τ=30-second leaderboard at N=15 would publish whichever of those three top-1s the random subsample favored, with no power to defend the choice against any of the others.

4.2 Sample Efficiency: The Right Test, Not the Right Scalar

The headline above leans on RMST and pooled logrank. We now ask the operational question: at fixed N, which metric most reliably detects a ranking difference? For each pair of inference models and each $N \in \{5, 10, 15, 20, 25, 30\}$ per (model, object) cell, we ran 300 outer subsampling trials. Each trial subsamples N episodes per cell with replacement, then runs a 200-rep inner episode-clustered bootstrap to compute a two-sided p-value for the difference of each candidate metric (macro-averaged across objects, RMST and KS computed at $\tau = 120\,\text{s}$ for the efficiency experiment). Detection rate is the fraction of trials with $p < 0.05$. Figure 1b shows the result for the borderline pair GR00T vs. ACT; Figure 4 shows the three pairs that carry efficiency information.

The table reads in three regimes. **Three saturated pairs** (any inference model vs. SmolVLA): all metrics work; effects are large; binary baselines are slightly faster than KS at saturation because the pooled-resample H_0 calibration is conservative when effects are huge. **Three close pairs** (the three top-3 model pairs) – the bucket where sample efficiency actually matters. Only GR00T vs. ACT reaches the 80% target within budget, on KS at N=20. KS detection rate at $N = 30$ on the other two pairs is 0.75 (OpenPI vs. ACT) and 0.61 (OpenPI vs. GR00T), while $F(30\,\text{s})$, $F(60\,\text{s})$, and RMST stay below 0.30 on every close pair. KS dominates the binary baselines by a factor of 3–6 in detection rate at $N \leq 30$ across the three close pairs; resolving them to 80% empirically would require the 600–1500 paired rollouts the analytical power calculation predicts (§3.3).

Detection rate at $N = 30$ averaged across all six pairs: $F(30\,\text{s}) = 0.54$, $F(60\,\text{s}) = 0.60$, RMST = 0.57, **KS** = 0.89. The +29 pp gain of KS over $F(60\,\text{s})$ is concentrated entirely on the three close pairs.

Two takeaways. First, **a CDF-level test (KS) is materially more sample-efficient than any**

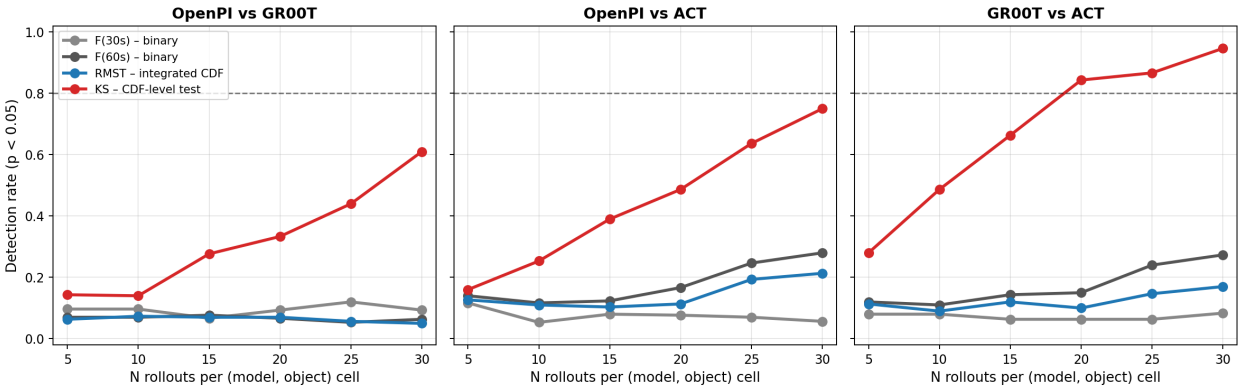


Figure 4: Detection rate vs. subsample size N per (model, object) cell, on the three close pairs that carry efficiency information (left: OpenPI vs. GR00T, the closest pair; centre and right: the two next-closest). Dashed line at 0.8 marks the conventional power target. KS (red) climbs steeply on every close pair while $F(30\,\text{s})$, $F(60\,\text{s})$, and RMST stay near the floor; KS reaches the 80% target within budget only on GR00T vs. ACT (right, N=20), climbing to 0.75 on OpenPI vs. ACT and 0.61 on OpenPI vs. GR00T at N=30 while the others stay below 0.30 across the entire range. The three any-vs-SmolVLA pairs saturate at $N = 5$ for every metric and are deferred to Appendix B. **The choice of test, not just the choice of N, determines what is resolvable.**

Pair	$F(30\,\text{s})$	$F(60\,\text{s})$	RMST	KS
OpenPI vs. GR00T	—	—	—	—
OpenPI vs. ACT	—	—	—	—
OpenPI vs. SmolVLA	5	5	5	10
GR00T vs. ACT	—	—	—	**20**
GR00T vs. SmolVLA	5	5	5	10
ACT vs. SmolVLA	5	5	5	10

Table 4: Smallest tested N at which detection rate ≥ 0.8 per (pair, metric). "—" means the detection rate did not reach 0.8 within the tested range $N \leq 30$. KS is the only metric that crosses the threshold within budget on a close pair (GR00T vs. ACT, N=20); on the other two close pairs KS reaches detection rates of 0.75 (OpenPI vs. ACT) and 0.61 (OpenPI vs. GR00T) at N=30, while every binary-threshold metric stays below 0.30.

single CDF-derived scalar (RMST, $F(60\,\text{s})$) for ranking. RMST is essentially tied with $F(60\,\text{s})$; the integration over $[0, \tau]$ does not buy ranking power, because the information lives in *where the curves separate* and an integrator smears that information. Second, **empirical and analytical evidence triangulate.** The closest pair is unresolvable at $N = 30$ by every metric, and the analytical 600–1500-paired-rollout requirement predicts exactly that. The protocol-design recommendation is direct: separate the reporting scalar from the ranking test. Report RMST or HRT as the headline; test rankings with paired KS (or stratified logrank) on the full CDF. Treating a single scalar as both the report and the test wastes statistical power.

What KS does and does not say. A two-sample KS test reports that two CDFs are different; it does not by itself say which policy is faster. The signed direction comes from a paired scalar – in our case the macro RMST difference (or, equivalently, the sign of $F_a(t) - F_b(t)$ at any t where the difference is large). In practice we read both: KS provides the existence claim ("these distributions differ at $p < 0.05$"), and the scalar provides the direction ("policy a has lower RMST than b"). The two outputs together are what we recommend reporting.

4.3 Aggregation-Rule Disagreement

The headline (Table 2) ranks under integrated RMST. We tested whether a different principled aggregation rule over the same CDF would agree. **Pairwise stochastic dominance vs. teleop**, AUC = $P(T_{\text{model}} < T_{\text{teleop}})$, macro-averaged across objects: ACT 0.138 [0.111, 0.167] > OpenPI 0.101 [0.084, 0.118] ≈ GR00T 0.096 [0.080, 0.112] > SmolVLA 0.027 [0.017, 0.038]. ACT is now *highest* among inference models. **Integrated metrics** (RMST, HRT, intervention rate): ACT third, behind GR00T and OpenPI.

The disagreement is not noise; it is structural, and on our data it is sharpened by a property of the teleop reference rather than by ACT alone. The teleop CDF is concentrated on small times – nearly all teleop attempts finish in under 30 s – so the AUC integral $P(T_{\text{model}} < T_{\text{teleop}})$ is dominated by what happens at the short-time end of the model's CDF. A policy with a fat "fast-when-it-works" early peak (ACT) wins more of those early-finish races against a tightly-distributed teleop than a steadily-paced one does, even if the steadily-paced policy completes more attempts overall. AUC asks "which policy reaches its first success first?"; RMST asks "which policy has the lower mean time-to-next-success including hard failures as ∞?". Both are well-defined; both are implemented in our public code; neither is "right."

Recommendation. We recommend **HRT as the default headline scalar**, because it inherits the joint reliability-throughput property of the CDF (via $T = \infty$ inflating the denominator) and matches operator-facing UPH practice. AUC against a much-faster reference is not a good co-headline on *this* dataset: as the data above show, when the reference's CDF is concentrated at small times the AUC integral is dominated by short-time behaviour, and "which model registered an early success" overrides "which model completed more attempts overall." This is a property of the teleop reference being seven times faster than any evaluated VLA, not a property of AUC itself; on a benchmark where the reference and the policies operated at comparable speeds, AUC and RMST would track much more closely. When the headline scalar disagrees with another principled aggregation, the right move is not to publish both with equal weight but to disclose which operational priority the choice encodes. Cross-model P-P plots (§4.4, without the teleop anchor) are useful as a visualization of distributional shape; they are a byproduct of the CDF, not an independent metric.

4.4 Pairwise Model-vs-Model Removes the Teleop-Anchor Bias

The teleop-anchored AUC of §4.3 saturates against teleop's fast CDF: for top-3 models nearly all model events fall in teleop's tail, so AUC is dominated by rare slow-teleop events. Pairwise model-vs-model P-P plots (without the teleop anchor) neutralize this. Macro-AUCs land near 0.5 for every top-3 pair, with 95% CIs that all cross 0.5: ACT-vs-OpenPI 0.51 [0.45, 0.56]; ACT-vs-GR00T 0.50 [0.45, 0.55]; OpenPI-vs-GR00T 0.50 [0.46, 0.54]. This confirms the RMST-side ranking that the top three frontier VLAs are pairwise indistinguishable on integrated metrics at our depth.

4.5 Per-Object Structure

Figure 7 shows per-(model, object) RMST with 95% episode-clustered bootstrap CIs. The cleanest visible pattern is the GR00T-vs-ACT inversion between Batteries (GR00T 89 s ≤ ACT 105 s; raw $p = 0.30$) and Scissors (ACT 99 s ≤ GR00T 116 s; raw $p = 0.24$). Neither clears Bonferroni at $\alpha = 0.05/24$ over the 24 per-cell pairwise tests. **This is itself a finding about benchmark sample-size requirements:** per-object structure is visible at our N but below the formal significance threshold, and the field's modal N could not even see the visible patterns. Per-cell RMST values and 95% CIs are tabulated in Appendix B.

PhAIL's protocol additionally logs the spatial configuration (camera/tote placement) with every episode; we observe a 22.2 pp completion-rate drop for GR00T when the external camera is on the opposite side from the outbound tote (full table in Appendix F). Effects of this size demonstrate that environmental confounds can move the measured ranking, and that protocol-level logging is a precondition for clean ranking claims.

5 Discussion

5.1 Position Against Prior Benchmarks

Simulated benchmarks (RoboCasa, CALVIN, ManiSkill, SIMPLER [RoboCasa, CALVIN, ManiSkill, SIMPLER]) are fast to iterate but do not capture real-world perception, latency, or safety. **Real-hardware research benchmarks** such as RoboArena [RoboArena] and RoboChallenge [RoboChallenge] use physical robots but compare via pairwise Elo or scalar success on household and lab tasks; neither reports throughput or reliability, nor compares distributionally over operations.

Competition-style evaluations (RGMC at ICRA/IROS, OCRTOC [OCRTOC], NIST Assembly Task Boards [NIST-ATB], the Real Robot Challenge [RealRobotChallenge], the Robothon Grand Challenge [So et al.], surveyed in [Sun et al.]) all force standardized scoring on a shared task. Closest in spirit is the Digital Robot Judge [So et al.], which builds task-centric performance databases via electronic task boards – complementary to our video-and-telemetry release. PhAIL differs in being live between events rather than episodic, anchoring the headline scalar to a teleoperated reference, and reporting time-to-success distributionally rather than as a per-attempt success rate.

Methodology critiques. Kress-Gazit et al. [Kress-Gazit et al.] and the LBM examination [Barreiros et al.] are existence proofs that good real-robot evaluation is feasible: blinded same-session A/B, hundreds of rollouts. They prescribe; we ship a metric framework with a reference implementation. Agarwal et al. [Agarwal et al.] are the closest precedent for distributional comparison in ML evaluation; we extend their approach to time-to-event data with $T = \infty$ ghost events.

5.2 Why CDF-Level Tests, Not Just CDF-Derived Scalars

The central new finding of this paper is that *which test you use determines what ranking you can resolve at a given N*, and that choosing the right test is a much larger lever than choosing the right scalar. RMST, our headline reporting scalar, is *not* measurably more sample-efficient than $F(60\,\text{s})$ for ranking. The integration step that makes RMST interpretable as "average time to next success" is the same step that smears the where-do-the-curves-separate information that a sup-norm test exploits. The result is the central methodological recommendation of the paper: scalars are for reporting, tests are for ranking; conflating them wastes power.

5.3 Honest Limits

Single embodiment (Franka FR3 + Robotiq 2F-85, fixed camera layout); generalization to mobile, bimanual, or different end-effectors is future work. Pick-and-place primitive only; the framework applies unchanged to any operation with a teleoperated reference and event-time outcome (insertion, packing, navigation, articulated manipulation), but PhAIL's empirical validation is restricted to bin-to-bin across four trained objects. $N \approx 26$ per cell is below what binary metrics need to rank close pairs (§4.2 treats this as a feature, not a bug). The empirical numbers are illustrative of what the methodology can resolve at our depth; per-model recipe optimization could shift them.

5.4 Recommendations

The single recommendation that does the most work, on top of everything above, is **run policies in blind, randomized rotation on the same fixture in the same session.** Operator and environment shift outcomes by margins comparable to or larger than gaps between adjacent models – the spatial-configuration sensitivity of Appendix F shows GR00T losing 22 pp completion when the camera and outbound tote are placed on opposite sides. If the two models being compared are not run interleaved on the same workspace state by the same operator on the same day, what looks like a model-vs-model effect can be an operator-vs-operator or session-vs-session effect. Same-fixture blinded random rotation is a precondition for the rest of the methodology to hold; without it, the CDF you measure is contaminated by the protocol that produced it.

The remaining recommendations follow from the body of the paper: (i) report the time-to-success CDF, not just a binary success rate at a single threshold; (ii) treat hard failures (items lost outside the workspace, items left dropped on the table at episode end, safety-stop terminations) as $T = \infty$ ghost events rather than

additional censoring; (iii) ground the headline scalar in a same-fixture teleoperated reference and macro-aggregate cell-wise across tasks; (iv) report scalars with paired episode-clustered bootstrap intervals, and test rankings with paired KS (or stratified logrank) on the full CDF; (v) plan N from the pair you most want to resolve, by plotting detection-rate-vs-N curves on pilot data (as in §4.2) rather than picking a round number.

5.5 Submissions Welcome

The leaderboard at `https://phail.ai` is open to submissions. Teams can run their own checkpoints on the published evaluation protocol (or on the PhAIL operator's hardware, on request) and have the resulting CDFs scored by the same framework against the same teleoperated reference. The four rankings reported in this paper reflect each model's default open-source recipe; we expect them to move as the field publishes better recipes, and the framework is built to absorb those updates.

5.6 Future Work

Four commitments follow from the open release.

API-based submissions. Teams expose an inference HTTP endpoint that takes the standard observation contract (camera images, end-effector pose, language prompt) and returns actions; PhAIL operators run rollouts on the shared hardware and score the resulting CDFs against the published teleoperated reference. Teams keep control of their weights and infrastructure; the rollout protocol stays consistent. This minimizes the engineering distance between "we have a VLA" and "we have a result on the leaderboard," which is the practical meaning of an open benchmark.

More tasks. Insertion (USB-A, Ethernet) and small-part assembly are the next PhAIL tasks. Both have unambiguous success states and admit a teleoperated reference. The methodology applies unchanged.

More architectures. As new public VLA releases appear, we evaluate them on the existing PhAIL tasks against the same teleoperated reference and update the leaderboard in place. The methodology is the constant; the rankings move as the field publishes.

Automated success detection. Manual placement annotation is the most expensive and most bias-prone step in the current pipeline. We are extending the detection pipeline to handle pick-and-place tasks without manual review, and to bootstrap directly on new tasks as they are added. This is what makes per-task evaluation cost tractable as the benchmark grows.

The evaluation framework is open source; teams can reproduce the full protocol in their own labs without coordination from us.

6 Conclusion

If the field adopts this methodology, three things change. **First, scalar choice becomes a disclosed methodological commitment, not a default.** A leaderboard that reports HRT discloses an integrated-throughput priority; one that reports AUC-vs-teleop discloses a pairwise-stochastic-dominance priority; the same data can support either ranking, and a paper that picks one without saying so has elided a load-bearing decision. **Second, ranking claims become falsifiable.** A KS test on the macro CDF with paired episode-clustered bootstrap and a published power calculation lets a reader check whether N was sufficient to support the claim being made; "no, we cannot rank these models at this N" becomes a publishable finding rather than a discounted footnote. **Third, the hidden cost of binary success-rate-at-threshold becomes visible:** on borderline policy pairs, the binary metric throws away enough information about the underlying CDF that no realistic N recovers it, while the corresponding CDF-level test resolves the same comparison at a tractable budget. The leaderboard at `https://phail.ai` is open to submissions, and we expect the methodology to be most useful as the field publishes better fine-tuning recipes and the rankings move accordingly.

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A Twelve-Paper Survey of Recent VLA Evaluation Practice

Table 5 surveys 12 recent real-robot VLA papers from 2023–2025. Modal per-condition N is 10–20; in 11 of 12 papers no confidence intervals or paired tests are reported. The single recent counter-example is the LBM examination [Barreiros et al.], which reports Wilson-score 95% CIs.

Paper	Per-condition N	Stats reported
RT-1 (RSS 2023)	per-task N not standardized; ∼3000 total	Point estimates only
RT-2 (CoRL 2023)	∼10–15 per task; ∼6000 total	Point estimates only
RT-X / Open X-Embodiment (ICRA 2024)	10–20 modal; up to 100/skill	Point estimates
OpenVLA (CoRL 2024)	∼10 real / 50 sim per task	No CIs
π_0 (Physical Intelligence 2024)	∼10 trials/task; partial credit	No CIs
$\pi_{0.6}$ (arXiv 2511.14759, 2025)	tens per task, variable	No CIs
GR00T N1 (NVIDIA 2025)	5 objects × 3 trials = 15/task real	Point estimates
ACT / ALOHA (RSS 2023)	∼25 trials/task	Point estimates
Mobile ALOHA (CoRL 2024)	20/task (5 for "Cook Shrimp")	Point estimates
DROID (RSS 2024)	10 rollouts per (task, method)	"Standard error" mentioned, no CIs
SmolVLA (HuggingFace 2025)	tens per real task	Point estimates
ECoT (CoRL 2024)	∼20/task (∼300 total / policy)	Point estimates
LBM Examination (Science Robotics 2026) [Barreiros et al.]	hundreds per condition	Wilson 95% CIs

Table 5: Per-condition N and statistical reporting in recent real-robot VLA evaluation papers. "Per-condition N" is the per-condition sample size for the modal task – the count of trials per (model, task) cell. "Stats" summarizes confidence-interval and paired-test reporting in the original paper.

Power arithmetic at modal N. At $N = 10$–20, the Wilson 95% CI on an observed 70% success rate is roughly $[0.40, 0.89]$ at N=10 or $[0.48, 0.86]$ at N=20. The minimum detectable difference between two unpaired proportions at 80% power, $\alpha = 0.05$ is ∼35 percentage points at N=20. Separating two top VLAs whose true gap is ∼5 pp requires ∼600–1500 paired rollouts (McNemar), or more under unpaired tests. The empirical detection-rate curves of §4.2 corroborate: at $N \leq 30$ paired per cell, no binary-threshold metric resolves the GR00T-vs-OpenPI ranking on our data.

Concurrent methodology critiques. Kress-Gazit et al. [Kress-Gazit et al.] explicitly argue that current real-robot evaluations are statistically anemic. STEP [Snyder et al.] and SureSim concur: "20–30 real-world trials are insufficient for statistically significant conclusions." These prescribe; we ship a metric framework with reference implementation that operationalizes the prescriptions on a shared station.

B Per-Cell %-of-Teleop and RMST Heatmaps

Per-(model, object) HRT heatmap; per-cell RMST grid with 95% bootstrap CIs; per-cell pairwise logrank p-values with Bonferroni and BH/FDR corrections.

C Methodological Framework D1–D10

Definitions of T, censoring rules, multi-grip handling, $T = \infty$ ghost events, KM and Greenwood derivations, RMST and KS estimators, episode-clustered bootstrap pseudocode. Full details in `FINDINGS.md`; abridged version in this appendix.

D Hardware Diagram and Camera Intrinsics

Hardware photo, fixture diagram, camera intrinsics, network and software stack.

E Fine-Tuning Dataset Composition

Object	Episodes	Duration (min)
Wooden spoons	167	∼340
Towels	112	∼178
Scissors	83	∼160
Batteries	88	∼134
Total	352	∼812

Table 6: Fine-tuning dataset (released alongside the benchmark, ∼13 hours of robot operation).

F Spatial Configuration Sensitivity

PhAIL's protocol logs the spatial configuration (camera and tote placement) with every episode. Between rollouts we vary outbound tote placement (left/right) and external camera position (left/right), creating four spatial configurations. Table 7 reports completion rate by camera/tote configuration. "Same-side" means the external camera and the outbound tote are on the same side of the workspace; "opposite-side" means they are on opposite sides, in which case the inbound tote partially occludes the outbound tote from the camera's perspective.

GR00T shows the largest sensitivity (22.2 pp drop), suggesting strong reliance on the external camera view; OpenPI is the most robust among the top models. An environmental change a human operator would

Model	Same-side	Opposite-side	Δ
OpenPI	51.1%	45.0%	+6.1 pp
GR00T	57.8%	35.6%	+22.2 pp
ACT	35.6%	31.6%	+4.0 pp
SmolVLA	12.1%	9.5%	+2.6 pp

Table 7: Completion rate by camera/tote configuration.

not notice can move the measured ranking. PhAIL's protocol logs the configuration with every episode so that this kind of confound is visible to the analyst rather than absorbed into noise.

G Power-Calculation Derivations

McNemar derivation for binary paired ranking; t-test derivation for RMST paired ranking; KS subsampling-based empirical power curves with full $(N, \text{metric}, \text{detection rate})$ tables.

H Annotation Protocol

The framework requires per-episode event logs (place timestamps, intervention counts, end-state item counts). The current protocol obtains them in two parallel streams; both are released. The protocol is implementation, not contribution, and is expected to evolve.

Crowd annotation. High-tier crowd annotators view the synchronized exterior and wrist videos at 30 fps via a custom Toloka task. They record the timestamp of each successful place, the count of items in the source and target totes at end of run, and a count of operator-visible interventions. Annotator instructions cover the three intervention classes (drops outside totes, safety stops, stalls) with edge-case examples drawn from a curated 10-episode pilot.

Autopilot. A telemetry parser emits per-event labels via three stages: *detect* extracts edges from the gripper command signal; *featurize* attaches per-edge features (hold duration, end-effector displacement, target-zone indicator, prior-open gap, post-close measured grip); *classify* maps the feature dict to a label (pick / place / drop / phantom). The classifier is the only stage that makes label decisions, isolating tunable thresholds in one file.

Cross-validation. Crowd labels are the gold reference. The autopilot is evaluated against them on every annotated episode; systematic disagreement is reported and used to refine the autopilot rather than override the gold. Both label streams are public.

I Reproducibility

Software environment (Python 3.13, `uv`, Linux), random seeds (0 throughout), expected runtimes (figure regeneration ∼10 minutes; sample-efficiency experiment ∼25 minutes), regeneration command (`bash build/paper_regen.sh`). Code at `https://phail.ai`.

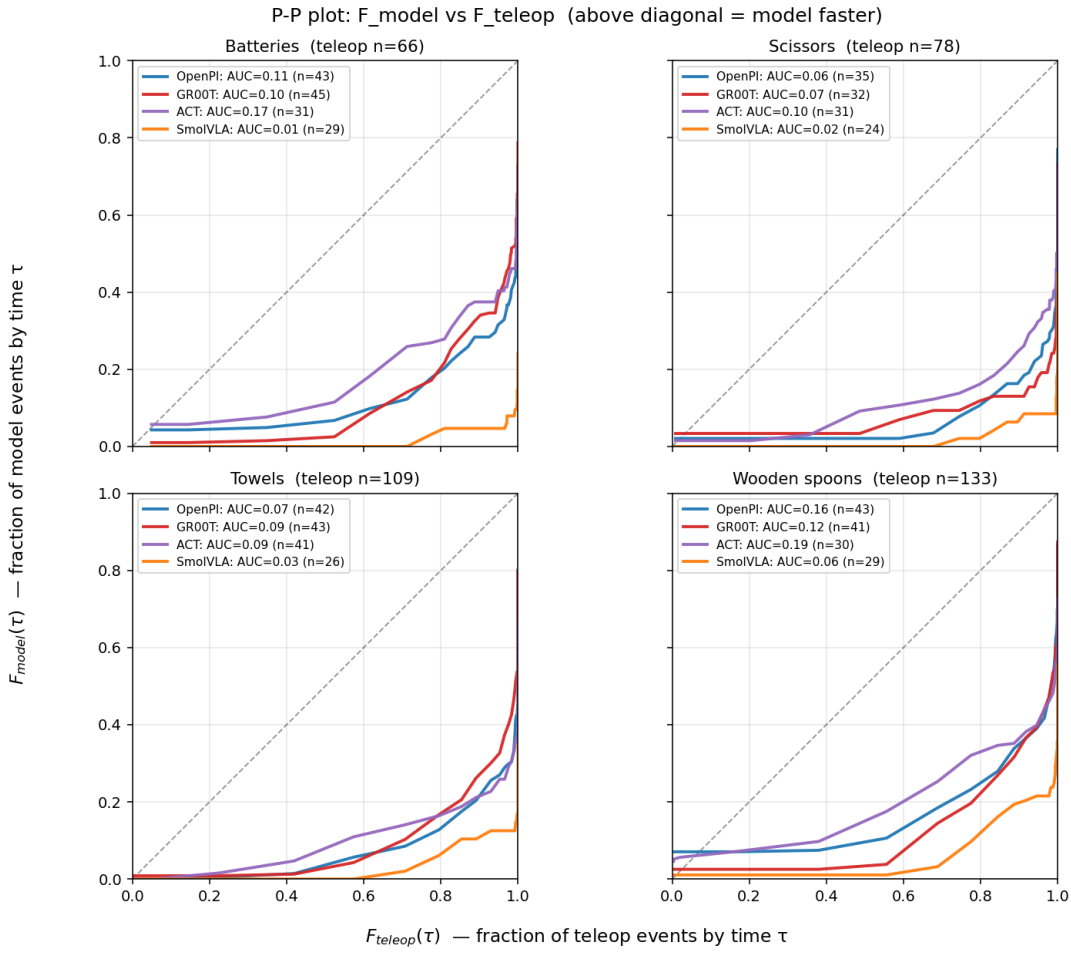


Figure 5: P-P plots versus the teleoperated reference, per (model, object) cell. Each curve plots $\big(F_{\text{teleop}}(t),\, F_{\text{model}}(t)\big)$ parametrically as t sweeps from 0 to 240 seconds; the diagonal is "as fast as the reference," below-diagonal is slower. The maximum vertical distance from the curve to the diagonal equals the Kolmogorov-Smirnov distance between the two CDFs; the area under the curve equals the AUC. The reference reaches $F = 1$ within $\sim$30 seconds, so the curves spend most of their length near the right edge: the only region where a model can score AUC points is the short-time corner near the origin, where ACT's early-success peak (visible as the jump near the y-axis on Batteries and Towels) lifts its score even though its integrated RMST places it third. The methodological point: the same CDF supports two principled aggregation rules with opposite rankings; reporting one scalar without disclosing the choice is structurally misleading.

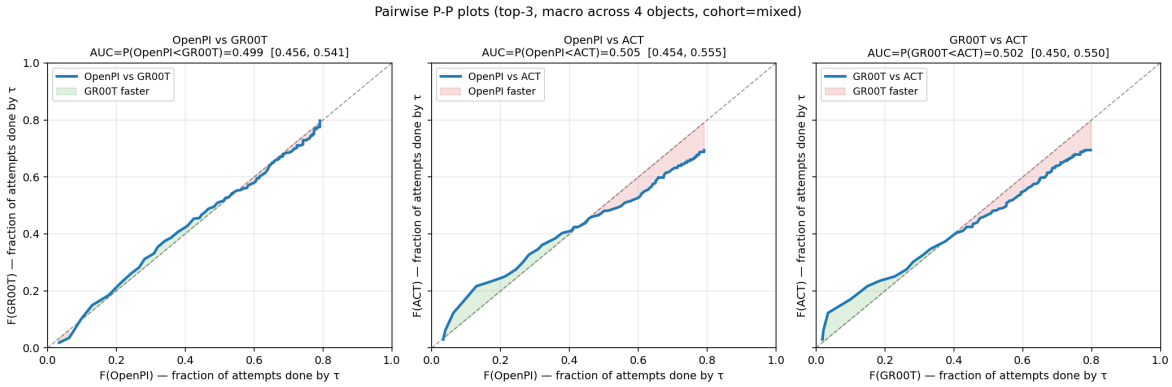


Figure 6: Pairwise model-vs-model P-P plots, top-3 only (SmolVLA excluded as too easily separable to be informative), macro-averaged across the 4 trained objects. Removing the teleop anchor neutralizes the saturation bias of Figure 5: every top-3 pair falls on or near the diagonal, with macro-AUCs all consistent with 0.5 (95% paired-bootstrap CIs cross 0.5 for every pair). **The top three frontier VLAs are pairwise indistinguishable at our depth on a comparison that does not depend on a fast external reference.**

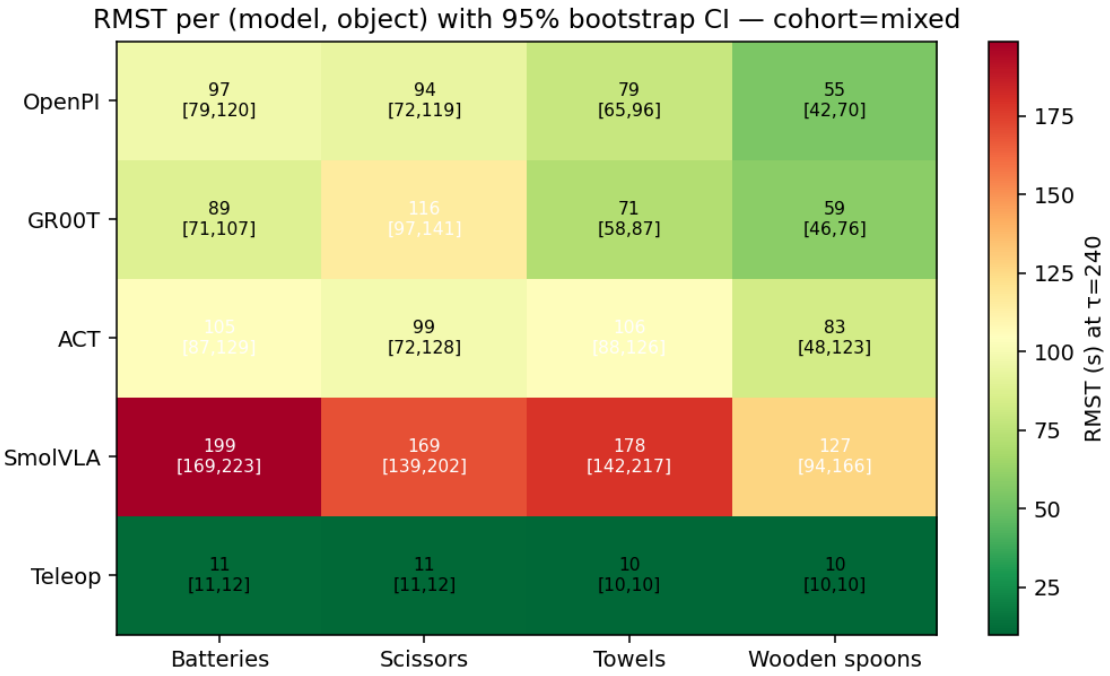


Figure 7: Per-(model, object) RMST with 95% episode-clustered bootstrap CIs. Crossings exist (most clearly GR00T vs. ACT between Batteries and Scissors) but none clears Bonferroni at $\alpha = 0.05/24$. The visible patterns suggest the rankings vary by object; resolving them formally requires the rollout budget that Figure 4 quantifies.